Compact Multi-Attribute Quality Assessment of Colonoscopy Videos

Department of Computer Science and Engineering
Shiv Nadar University Chennai, India

***Abstract*—Colonoscopy videos are often degraded by motion blur, poor lighting, specular glare and fluid or debris, and such degradations make a clip harder to interpret. An automatic quality score could warn the endoscopist while the examination is still running. We study this problem on the MAC-VQA database, which holds 1,000 clips rated by 15 observers on five distortion attributes and on overall quality. Our objective is to predict all six mean opinion scores with one compact model. The network works on frozen ResNet-50 features: it scores every image patch for each attribute, weights patches and frames with learned attention, and combines them with a small temporal Transformer, a ranking loss and hand-crafted photometric descriptors, joining its two spatial branches by plain concatenation. Over 20 random 720/80/200 splits, taking the ten best iterations by overall test SRCC, it reaches an overall Spearman rank-order correlation (SRCC) of 0.945±0.004, against 0.940 reported for the published state of the art, DARNet, while trailing it on the other scores (six-score mean SRCC 0.880 versus 0.887). A study of 34 variants shows that photometric descriptors and the number of frames matter, whereas attention and frame order add little. Code repository: https://github.com/your-username/your-project**

***Index Terms*—Colonoscopy, video quality assessment, multi-attribute quality, patch attention, Transformer, SRCC and PLCC, MAC-VQA**

I. Introduction

Colorectal cancer is among the most lethal cancers worldwide [1], [2], and colonoscopy is the main tool for finding and removing early lesions. In practice the recorded video is rarely clean. The camera moves quickly, the bowel is unevenly lit, the wet mucosa reflects the light source, and bubbles or stool cover parts of the wall [3]. A clip with these problems can hide a lesion, so a fast and objective quality score would let the operator re-inspect a segment or discard a poor recording before it reaches a diagnostic or computer-aided system.

Most video quality assessment (VQA) research targets natural videos with synthetic or streaming distortions, and the few medical studies that exist rely on synthetic distortions or do not give attribute-level opinion scores [4], [5]. Yue *et al.* closed part of this gap with MAC-VQA, a database of 1,000 authentically degraded colonoscopy clips rated on five distortion attributes (temporal-spatial visibility, brightness, specular reflection, stability and utility) and on overall quality, together with DARNet, a network that reaches state-of-the-art accuracy on it [3].

Scope of this work. The project is paper-based but it is an extension study, not a reproduction. We did not re-implement or re-train DARNet; its numbers in this report are the published ones. We ask a narrower question: how much of that accuracy can a small model recover when it is built on frozen ResNet-50 features, with explicit patch-level and frame-level reasoning per attribute, and which of its parts actually matter? DARNet uses a ResNet-50 and a Video Swin Transformer in parallel, cross-attention between attributes, and a graph network for the overall score, which is a lot of machinery for a database of only 1,000 clips.

Our contributions are the following.

- A compact network (4.26 M trainable parameters, frozen backbone) that scores patches per attribute, aggregates them with learned spatial and temporal attention, and predicts all six scores. Its two spatial branches are fused by plain concatenation (Section III-B).
- An evaluation on 20 random splits, reporting the ten best iterations (selected on the test result, and stated as such), with the logistic mapping fitted on validation data only, compared with DARNet on all six scores and four metrics.
- A 34-variant ablation, with a paired significance test, showing which components carry the performance and which do not.

Section II reviews related work, Section III describes the data and the model, Section IV reports and discusses the results, and Section V concludes.

II. Related Works

Classical and deep VQA for natural videos. Early no-reference methods describe each video with hand-designed statistics and regress the score; TLVQM [6] and VIDEVAL [7] are the usual examples. Deep models replaced them: VSFA [8] feeds frame features from a pretrained CNN to a recurrent unit with temporal pooling, Simple-VQA [9] adds quality-aware pretraining, and BVQA [10] transfers knowledge from two source domains and models motion. Transformer-based methods followed, notably FAST-VQA and FasterVQA [11], [12], which sample small "fragments" to keep a video Transformer affordable, and DOVER [13], which combines an aesthetic and a technical branch. These methods predict a single overall score and were designed for user-generated content, so they cannot report which distortion hurts a colonoscopy clip.

Medical video quality. Lévêque *et al.* studied telesurgery videos with synthetic transmission errors [4], and Ali *et al.* proposed a framework for quality assessment and restoration in endoscopy video [5]; neither provides attribute-level opinion scores for authentic colonoscopy distortions. For colonoscopy images, Yue *et al.* built a perceptual benchmark of enhanced images and an objective model [14].

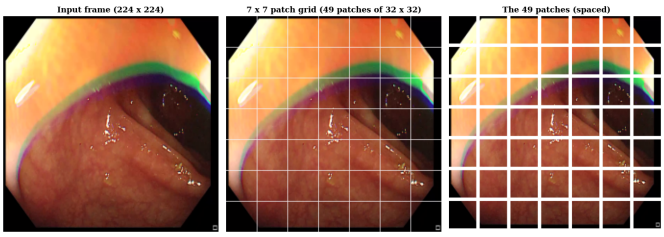


Fig. 1. Model input for one clip (8-3): a sampled frame, its 7×7 patch grid and the 49 patches.

DARNet. The source paper [3] introduced MAC-VQA and DARNet. Its spatio-temporal feature unit runs a frozen ResNet-50 on 16 frames and a Video Swin-T [15] on stitched fragments, a multi-attribute association module applies cross-attention between the five attribute streams, and a behaviour-guided reasoning module (a two-layer graph network over attributes whose edges come from the correlation of the human ratings) predicts the overall score. It reports an overall SRCC of 0.940 and is superior to nine other VQA methods. Its limits are cost (two large backbones), no study of which parts matter, and no patient-independent evaluation. Our model keeps attribute-wise outputs and a learned overall head but replaces the heavy streams with frozen features, patch-level scoring and light attention.

III. Proposed Methodology

A. Problem Formulation and Dataset

The input is a colonoscopy clip of 9–11 seconds; the output is a vector $\hat{\mathbf{y}} \in \mathbb{R}^6$ with the predicted mean opinion scores (MOS) of temporal-spatial visibility (TSV), brightness (B), specular reflection (SR), stability (S), utility (U), and overall quality (O). MAC-VQA [3] contains 1,000 clips cut from 40 full-length colonoscopies recorded at one hospital with an Olympus CV290, rated by 15 observers. After outlier screening and z-scoring the ratings are averaged into one MOS per clip and score; attribute scores lie in $[0, 3]$ and overall quality in $[1, 5]$, higher being better. Only these averaged MOS are available to us, not the individual ratings.

All 1,000 videos decode and match the 1,000 MOS records one to one, with no missing values and no duplicate identifiers. The clips are about 1222×1080 pixels at 24.3 fps and last 10.5 s on average (9.6–11.5 s); clip length is unrelated to quality ($|\text{SRCC}| \leq 0.03$). The attribute scores have means of 1.55–1.74 (std. 0.66–0.80) on the 0–3 scale and use almost the whole range; the overall score has mean 3.08 and std. 1.01 on the 1–5 scale, and all distributions are slightly skewed (-0.45 to $+0.18$). The five attributes correlate strongly with the overall score (Spearman 0.91 TSV, 0.90 S, 0.92 U, 0.76 SR, 0.63 B), and brightness is the least related to the others (Fig. 4(c)). From each clip we take 16 frames at evenly spaced positions and resize them to 224×224; each frame is later divided into a 7×7 grid of 49 patches of 32×32 pixels (Fig. 1).

In every iteration, clips are split at random into 720 training, 80 validation and 200 test clips; iteration s uses random seed s for both the split and the training, and we run 20 iterations ($s = 0, \ldots, 19$) and report the ten with the highest overall test SRCC (iterations 1, 2, 3, 5, 9, 10, 12, 15, 16 and 17). This selection uses the test labels, so the reported numbers are optimistic (see Section IV-C). The validation set is used for early stopping and for fitting the output mapping, and the test set is evaluated once per iteration, after training. Leakage precautions: features are extracted with a frozen network, so no label information enters them; descriptor normalisation statistics and the target min–max ranges come from the training clips only; and the logistic mapping is fitted on validation predictions only. A random split, like DARNet's, can put clips of the same recording on both sides, so we also report a group-held-out split (Section IV-B). Fig. 2 shows why: the 72 file-name groups hold 13.9 clips on average, the group explains 24–40% of the variance of each score (η^2), and in 99.9% of the test clips of a random 720/80/200 split another clip of the same group is in the training set. Nine simple photometric measurements on 400 random clips (Fig. 2(a)) show that sharpness correlates best with the overall score (0.49) and with S (0.53), and dark-pixel fraction with B (-0.36), supporting their use as side input.

B. Processing Pipeline and Approach

Figure 3 gives the overview. **Preprocessing and features.** Sixteen frames are sampled uniformly from each clip and resized to 224×224. A torchvision ResNet-50 with ImageNet weights [16] stays frozen. From it we keep the layer-4 map (7×7×2048) and the layer-2 and layer-3 maps, average-pooled to 7×7 (512 and 1024 channels), so that each of the N=49 patches of frame t has a token $P_{t,i} \in \mathbb{R}^{3584}$ that mixes low-level and semantic information. Each frame also gets a 16-value photometric descriptor d_t: mean, standard deviation and 5th/95th percentiles of luminance, the fractions of very dark, very bright and specular-like pixels, mean and spread of saturation, a colourfulness measure, Laplacian variance, mean gradient magnitude, edge density, a noise estimate, a red/green ratio and the histogram entropy. These cues are cheap and directly related to brightness, glare and blur. All features are computed once and cached.

Spatial stage. Tokens are projected to D=256 ($x_{t,i} = \phi(P_{t,i})$) and passed through one Transformer encoder layer with learned position embeddings to get context tokens $c_{t,\cdot} = \text{Enc}(x_{t,\cdot} + e)$. The two are fused by plain concatenation, $f_{t,i} = \text{GELU}(\text{LN}(W[x_{t,i}; c_{t,i}]))$. For each of the five attributes a, two small heads read $f_{t,i}$: a patch quality $q^a_{t,i} = \sigma(h^a_q(f_{t,i}))$ and a spatial criticality $\alpha^a_{t,i} = \text{softmax}_i\, h^a_s(f_{t,i})$. The frame quality is

$$Q^a_t = \sum_{i=1}^{N} \alpha^a_{t,i}\, q^a_{t,i}. \tag{1}$$

Temporal stage. A frame context $g_t = \psi(\bar{P}_t) + \text{MLP}(d_t)$ is built from the mean patch token $\bar{P}_t$ and the photometric descriptors, fused with the five frame qualities, and fed to a two-layer temporal Transformer, giving z_t. Attribute-specific

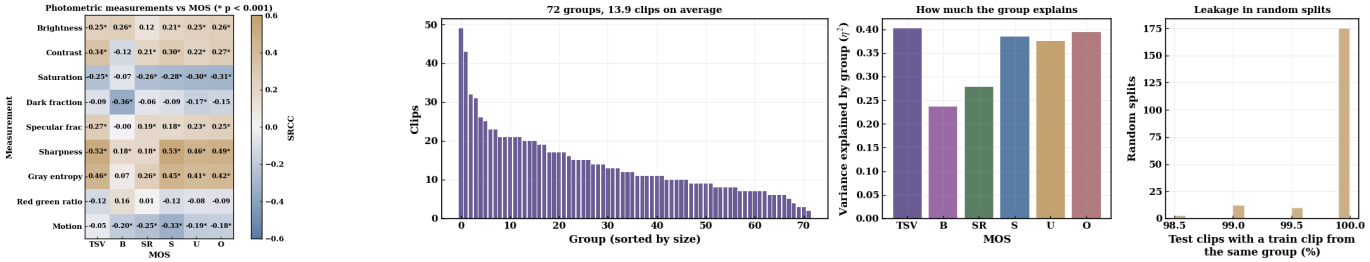


Fig. 2. Data analysis. (a) Spearman correlation between nine photometric measurements and the six MOS (* $p < 0.001$). (b) Left to right: clips per file-name group; variance of each score explained by the group; share of test clips with a same-group clip in the training set, over 200 random splits.

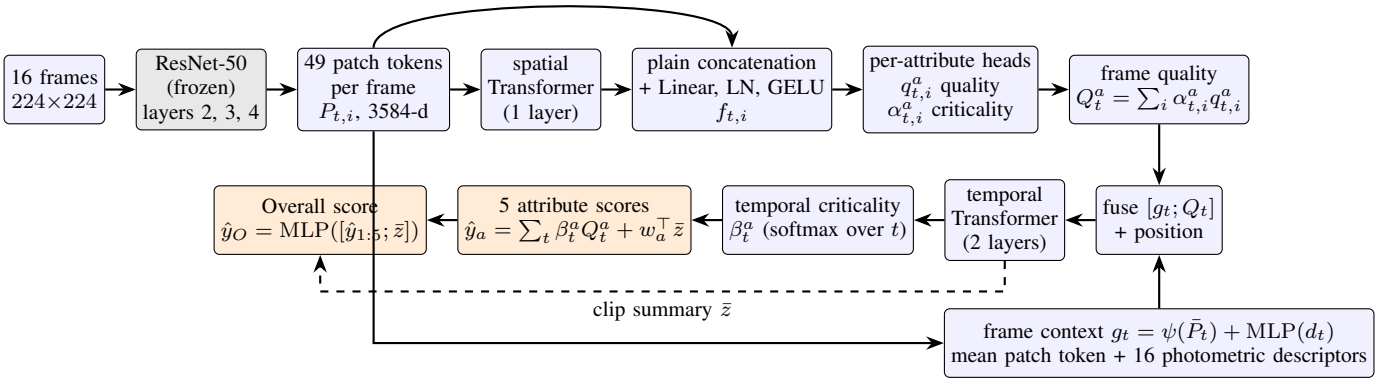


Fig. 3. Architecture of the final model. Frozen ResNet-50 features (layer-4 map plus pooled layer-2/3 maps) give 49 patch tokens per frame. A one-layer spatial Transformer produces a context token for each patch; the two tokens are joined by *plain concatenation* (instead of an adaptive gate). Per-attribute heads give a patch quality q and a spatial criticality α, whose weighted sum is the frame quality Q_t. After a temporal Transformer, per-attribute temporal criticality β pools the frame qualities, a global residual (res.) is added, and a learned head predicts the overall score from the five attribute scores and the clip summary $\bar{z}$.

temporal criticality $\beta^a_t = \mathrm{softmax}_t\, h^a_t(z_t)$ pools the frame qualities into the clip score

$$\hat{y}_a = \sum_{t=1}^{16} \beta^a_t\, Q^a_t + w_a^\top \bar{z}, \tag{2}$$

where $\bar{z}$ is the clip summary (mean of z_t plus a photometric context vector) and w_a is a linear residual initialised to zero. The overall score uses a learned head instead of a fixed formula: $\hat{y}_O = \mathrm{MLP}([\hat{y}_1, \ldots, \hat{y}_5; \bar{z}])$.

Training and prediction. Targets are min–max scaled with training statistics. The loss is the mean squared error over the six scores plus a pairwise ranking hinge,

$$\mathcal{L} = \tfrac{1}{6}\|\hat{\mathbf{y}} - \mathbf{y}\|^2 + \lambda_r \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \max\big(0,\, m - (\hat{y}_i - \hat{y}_j)\big), \tag{3}$$

with $\mathcal{P}$ the pairs in a batch whose true scores differ by more than 0.05 (for every one of the six scores), m=0.05 and λ_r=0.1. At test time a clip's cached features go through the network once. Predictions are mapped to the MOS scale by the four-parameter logistic function used in VQA [17],

$$f(s) = \frac{\eta_1 - \eta_2}{1 + \exp\big((\eta_3 - s)/|\eta_4|\big)} + \eta_2, \tag{4}$$

whose parameters are fitted on the validation predictions of that iteration and applied unchanged to the test set. **Differences from DARNet:** no video Transformer, no cross-attention between attributes and no graph module; the ResNet is never fine-tuned; patch-level scoring with spatial and temporal attention, photometric descriptors and a ranking loss are added. Each of these ideas is standard on its own, and we do not claim that the model is new in its parts.

C. Implementation Details

The code uses Python, PyTorch 2.11, OpenCV and SciPy and ran on one Tesla T4 GPU on Kaggle. Features are extracted once (10–15 min) and cached. We use AdamW [18] with learning rate 5×10^{-4}, weight decay 0.01, two warm-up epochs followed by cosine decay, gradient clipping at 1.0, batch size 32, dropout 0.2, 4 attention heads and at most 40 epochs. Training stops early when the validation mean SRCC has not improved for 10 epochs, and the best-validation weights are kept; on average the best epoch was 10.6 of 20.6 run. The model has 4,260,245 trainable parameters. One training run takes 62 s on average and inference takes 1.5 ms per clip from cached features (the one-time ResNet pass is not included). The repository will contain the notebooks, split seeds and per-run results; no patient data is published.

IV. Results and Discussion

A. Experimental Setup and Evaluation

We report four standard metrics per score. SRCC and Kendall's rank correlation (KRCC) measure monotonic agreement with the MOS and do not depend on the output scale. The

Pearson linear correlation (PLCC) and the root-mean-square error (RMSE) measure accuracy, and are computed after the validation-fitted logistic mapping. Higher is better for the first three and lower for RMSE. All numbers below are held-out test results; validation data were used only for early stopping and mapping. Values are mean ± sample standard deviation over iterations. The DARNet numbers are copied from its paper, which used an 8:2 split without a validation set and 10 trials; we did not re-run DARNet, so the comparison is indicative, not paired.

B. Quantitative and Qualitative Results

Table I compares the final model with DARNet. On overall quality our SRCC is 0.945 ± 0.004 against 0.940, and PLCC (0.944 against 0.943) and KRCC (0.796 against 0.790) are also slightly higher; the margins are small, of the order of one standard deviation, so we read them as a tie in favour of our model rather than a clear win, particularly since the ten iterations were selected on the test result. On the other scores DARNet is ahead: the SRCC gaps are 0.004 (TSV), 0.021 (B), 0.011 (SR), 0.005 (S) and 0.006 (U), and each is smaller than one standard deviation of our spread over iterations. The gaps are clearer for PLCC and RMSE (overall RMSE 0.344 against 0.332). Averaged over the six scores, our SRCC is 0.880 against 0.887. We therefore describe the model as matching or slightly exceeding DARNet on overall quality and trailing it slightly on the individual attributes.

Table II places the result among the nine methods benchmarked on MAC-VQA in [3]. Our overall SRCC (0.945), PLCC (0.944) and KRCC (0.796) are higher than those of all nine, including DARNet, and our RMSE (0.344) is second only to DARNet's 0.332. Because those numbers were produced by the DARNet authors under their own split, and ours are the best ten of 20 iterations, this ranking is a guide, not a controlled comparison.

Figure 4 and Fig. 5 show the behaviour behind these numbers. The overall scatter (Fig. 5, last panel) is tight and almost linear, with a few clips predicted too low at the high end. Brightness is the hardest dimension for both methods: its SRCC is lowest (0.777), its spread over iterations is the largest (std 0.037) and its scatter is the widest. The correlation matrix explains part of this. Brightness correlates only 0.48–0.61 with the other scores, while TSV, S and U correlate 0.84–0.92 with each other and with the overall MOS, so brightness cannot borrow strength from related targets, and its labels are less consistent. Specular reflection (SRCC 0.830) is the second hardest, probably because glare covers a small, fast-changing part of the frame that a 7×7 patch grid describes coarsely.

Attribute correlations in the predictions. Fig. 6 shows the correlation between the six predicted scores in each of the ten reported iterations. The predicted scores are more strongly correlated with each other than the MOS are (e.g., TSV–S 0.92–0.96 against 0.87–0.90 in the MOS, B–TSV 0.50–0.68 against 0.43–0.52), i.e., the model shares more information between attributes than the observers did.

Held-out groups. Clips whose file names share a prefix (72 groups of about 14 clips, which we believe to be recordings but could not confirm as patients) were kept together in one partition. On 10 such splits the overall SRCC drops from 0.940 to 0.923 ± 0.018 and the six-score mean SRCC from 0.880 to 0.855 ± 0.015. Random splits therefore overstate accuracy for new recordings; the same holds for the published protocol.

Ablation. Table III lists a selection of the 34 variants (all in the repository). Each variant changes one thing in the final model and is trained on the same ten splits; the final model has Mean(6) SRCC 0.8795 on these splits. Four findings stand out. (i) Photometric descriptors help: removing them costs 0.007 (p=0.006), and shuffling them across clips, which keeps the parameters but removes the information, costs more (0.010), so the gain comes from the content and not from extra capacity. (ii) Sampling matters: using 8 frames instead of 16 costs 0.026, the largest loss, and taking the maximum over frames instead of a weighted sum costs 0.012. (iii) The attention modules do not help measurably: removing spatial attention, temporal attention or the temporal Transformer changes Mean(6) SRCC by at most 0.002 with $p > 0.2$, and removing all four parts together (0.008 lower) behaves like removing the photometric descriptors alone. (iv) Replacing the concatenation with the adaptive gate used in our earlier version gives 0.8790 versus 0.8795 (p=0.23), so concatenation is a simplification, not an improvement. Coarser patch grids, a CNN-only spatial branch and a temporal-consistency loss each score about 0.006 higher, but with ten splits the smallest attainable p-value is 0.002, and after a correction for 34 tests none of the positive differences is significant, so we treat them as leads, not as findings.

C. Discussion and Limitations

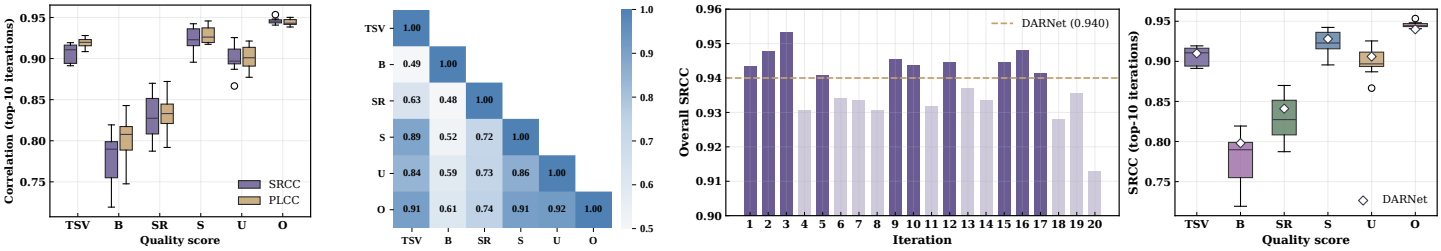
The ablation and two diagnostics suggest that the model works mainly as a strong frozen-feature regressor with useful photometric side information. The learned spatial attention is almost uniform (normalised entropy 0.972), the temporal attention is even flatter (0.993) and the five attribute-specific temporal curves differ little (mean L_1 distance 0.09). Shuffling the order of the test frames changes the overall SRCC from 0.9453 to 0.9449, so the model hardly uses temporal order; this is a limit of the design, because motion-related distortions such as stability are then judged from per-frame appearance, and 720 training clips may be too few to learn motion cues.

The main limitations are as follows. First, the reported iterations are the ten best of 20 by their own test SRCC, and the final configuration was chosen after looking at test-split results of earlier variants, so the absolute numbers are optimistic; the mean of all 20 iterations is lower (overall SRCC 0.938 ± 0.009, six-score mean 0.878) and ties DARNet within noise. Second, the comparison with DARNet uses published numbers from a different protocol (8:2 split, 10 trials, no validation set), not a re-run. Third, random splits can leak recording-specific appearance, and our groups may not be patients. Fourth, only averaged MOS were available, so DARNet's per-observer analyses could not be repeated, and inference speed excludes the ResNet pass.

TABLE I

RESULTS ON MAC-VQA, TEST SET. DARNET: PUBLISHED VALUES (10 TRIALS). OURS: FINAL MODEL, MEAN ± STD OVER THE 10 ITERATIONS WITH THE HIGHEST OVERALL TEST SRCC, OUT OF 20 RANDOM SPLITS (*SELECTED ON THE TEST RESULT, HENCE OPTIMISTIC). BOLD MARKS THE BETTER VALUE OF THE PAIR. MEAN(6) IS THE AVERAGE OVER THE SIX SCORES (FOR DARNET, THE AVERAGE OF ITS SIX REPORTED SCORES).

	DARNet [3]				Ours (M9 + plain concat.), top-10 iterations*			
Score	SRCC↑	PLCC↑	KRCC↑	RMSE↓	SRCC↑	PLCC↑	KRCC↑	RMSE↓
TSV	**0.910**	**0.924**	**0.744**	**0.252**	0.906±0.012	0.919±0.006	0.740±0.014	0.262±0.012
B	**0.798**	**0.821**	**0.607**	**0.459**	0.777±0.037	0.801±0.032	0.588±0.035	0.494±0.036
SR	**0.841**	**0.855**	**0.654**	**0.343**	0.830±0.028	0.832±0.025	0.641±0.031	0.388±0.027
S	**0.928**	**0.933**	**0.768**	**0.276**	0.923±0.015	0.929±0.011	0.761±0.020	0.295±0.019
U	**0.906**	**0.914**	**0.737**	**0.283**	0.900±0.017	0.901±0.015	0.727±0.023	0.317±0.020
Overall	0.940	0.943	0.790	**0.332**	**0.945**±0.004	**0.944**±0.004	**0.796**±0.008	0.344±0.010
Mean(6)	**0.887**	**0.898**	**0.717**	**0.324**	0.880±0.011	0.888±0.010	0.709±0.012	0.350±0.011



(a) SRCC and PLCC, top-10 (b) PLCC between the MOS (c) Top-10 iterations: overall SRCC; SRCC per score

Fig. 4. (a) Spread of SRCC and PLCC over the top-10 iterations for each quality dimension. (b) Correlation matrix of the six MOS over all 1,000 clips. (c) Left: overall SRCC per iteration (top-10 dark, dashed line = DARNet); right: SRCC of the six scores over the top-10 iterations (boxes) and DARNet (diamonds).

TABLE II

OVERALL QUALITY ON MAC-VQA: PUBLISHED RESULTS OF NINE VQA METHODS (QUOTED FROM [3], PROTOCOL OF THAT PAPER) NEXT TO OURS (MEAN OF THE 10 BEST OF 20 ITERATIONS; *CHOSEN BY TEST RESULT, HENCE OPTIMISTIC).

Method	SRCC↑	PLCC↑	KRCC↑	RMSE↓
TLVQM [6]	0.887	0.885	0.702	0.466
VIDEVAL [7]	0.888	0.891	0.709	0.461
VSFA [8]	0.932	0.934	0.775	0.356
Simple-VQA [9]	0.897	0.901	0.718	0.433
BVQA [10]	0.928	0.936	0.769	0.354
FAST-VQA-M [11]	0.898	0.902	0.721	0.433
FASTER-VQA [12]	0.930	0.930	0.771	0.369
DOVER [13]	0.933	0.936	0.778	0.352
Light-VQA [19]	0.836	0.842	0.643	0.550
DARNet [3]	0.940	0.943	0.790	**0.332**
Ours (top-10 iterations*)	**0.945**	**0.944**	**0.796**	0.344

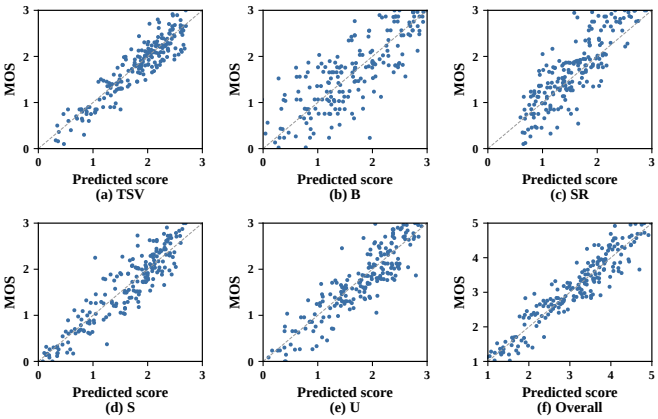


Fig. 5. Test-set scatter plots of the six scores for one iteration (iteration 1, one of the top ten): prediction after the validation-fitted mapping against MOS; dashed line = ideal.

V. CONCLUSION AND FUTURE WORK

We built a compact multi-attribute quality model for colonoscopy videos on frozen ResNet-50 features, with patch-level scoring, a temporal Transformer, photometric descriptors and a ranking loss, and evaluated it on MAC-VQA with 20 random splits and a validation-fitted logistic mapping. Over the ten best iterations its overall SRCC is 0.945 ± 0.004, slightly above DARNet's 0.940, but it is behind DARNet on the five attribute scores and in RMSE; since the iterations were selected on the test result, the margin should be read with care. The ablation indicates that photometric information and a sufficient number of frames are what count here, while attention and frame order contribute little, and concatenation works as well as a gate. A group-held-out split lowers the overall SRCC by about 0.017, which shows that random-split numbers should be read with care.

Natural next steps are to verify the patient identity of each clip and evaluate on truly patient-independent folds, to re-test the coarser-grid and CNN-only variants on more splits with validation-only selection, and to add a lightweight motion cue (frame differences or optical flow), since the current model ignores frame order and stability is a motion attribute.

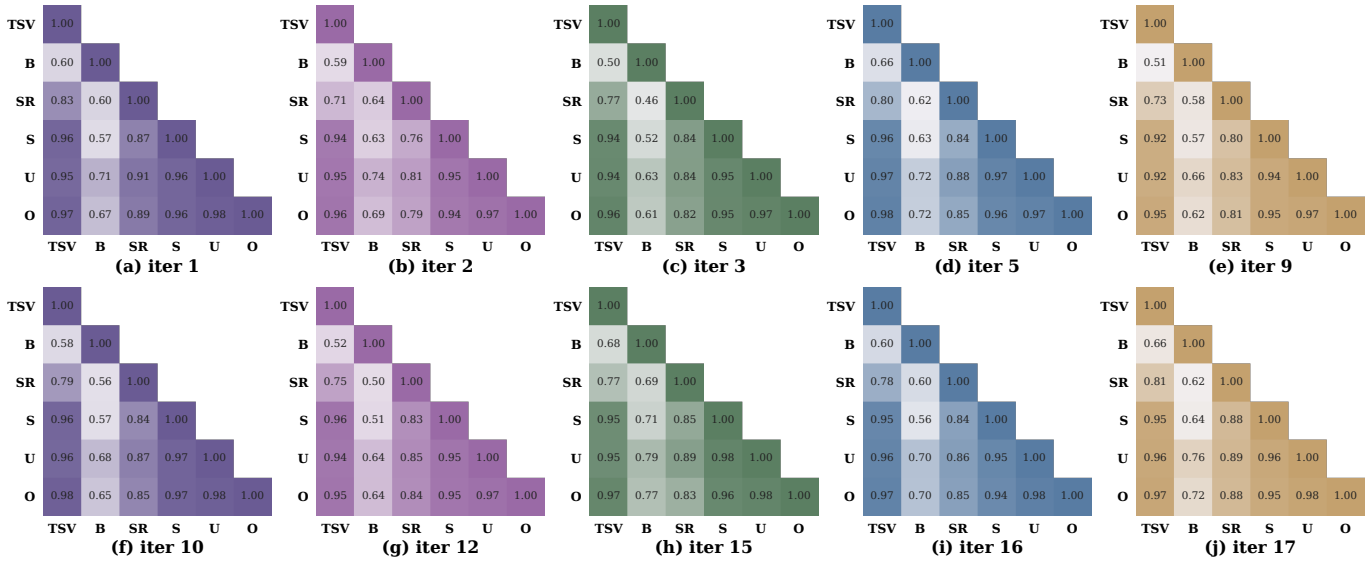


Fig. 6. Pearson correlation between the six predicted scores (raw test predictions) in each of the top-10 iterations; only the lower triangle is shown.

TABLE III
ABLATION OF THE FINAL MODEL ON 10 SPLITS (SEEDS 0–9). Δ IS THE CHANGE OF MEAN(6) SRCC RELATIVE TO THE FINAL MODEL; p IS A PAIRED WILCOXON TEST OVER THE 10 SPLITS, UNCORRECTED FOR THE 34 COMPARISONS.

Variant	Mean(6) SRCC	O SRCC	Δ	p
Final model (all parts on)	0.8795	0.9403	–	–
− photometric descriptors	0.8724	0.9376	-0.0071	0.006
− multi-stage features	0.8752	0.9330	-0.0043	0.160
− temporal attention	0.8789	0.9378	-0.0006	0.846
− spatial attention	0.8778	0.9386	-0.0017	0.232
− all four parts	0.8717	0.9300	-0.0078	0.006
Photometric descriptors shuffled	0.8694	0.9325	-0.0101	0.002
8 frames instead of 16	0.8539	0.9206	-0.0256	0.002
Temporal max-pooling	0.8677	0.9352	-0.0118	0.002
Spatial max-pooling	0.8807	0.9400	+0.0012	0.322
Adaptive gate instead of concat.	0.8790	0.9372	-0.0005	0.232
CNN branch only (no spatial Transf.)	0.8852	0.9391	+0.0057	0.004
2×2 patch grid	0.8855	0.9413	+0.0060	0.131
3×3 patch grid	0.8853	0.9387	+0.0058	0.004
+ consistency loss (λ=0.5)	0.8857	0.9393	+0.0062	0.037
No ranking loss	0.8758	0.9375	-0.0037	0.160
No temporal Transformer	0.8801	0.9390	+0.0006	0.846
Group-held-out split (fin. model)	0.8547	0.9227	-0.0248	–

REFERENCES

[1] H. Sung *et al.*, "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA Cancer J. Clin.*, vol. 71, no. 3, pp. 209–249, Feb. 2021.

[2] N. Keum and E. Giovannucci, "Global burden of colorectal cancer: Emerging trends, risk factors and prevention strategies," *Nat. Rev. Gastroenterol. Hepatol.*, vol. 16, no. 12, pp. 713–732, Dec. 2019.

[3] G. Yue, L. Zhang, J. Du, T. Zhou, W. Zhou, and W. Lin, "Subjective and objective quality assessment of colonoscopy videos," *IEEE Trans. Med. Imag.*, vol. 44, no. 2, pp. 841–854, Feb. 2025, doi: 10.1109/TMI.2024.3461737. Code and database: https://github.com/cheunglaihip/DARNet.

[4] L. Lévêque, W. Zhang, C. Cavaro-Ménard, P. Le Callet, and H. Liu, "Study of video quality assessment for telesurgery," *IEEE Access*, vol. 5, pp. 9990–9999, 2017.

[5] S. Ali *et al.*, "A deep learning framework for quality assessment and restoration in video endoscopy," *Med. Image Anal.*, vol. 68, Art. no. 101900, Feb. 2021.

[6] J. Korhonen, "Two-level approach for no-reference consumer video quality assessment," *IEEE Trans. Image Process.*, vol. 28, no. 12, pp. 5923–5938, Dec. 2019.

[7] Z. Tu, Y. Wang, N. Birkbeck, B. Adsumilli, and A. C. Bovik, "UGC-VQA: Benchmarking blind video quality assessment for user generated content," *IEEE Trans. Image Process.*, vol. 30, pp. 4449–4464, 2021.

[8] D. Li, T. Jiang, and M. Jiang, "Quality assessment of in-the-wild videos," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 2351–2359.

[9] W. Sun, X. Min, W. Lu, and G. Zhai, "A deep learning based no-reference quality assessment model for UGC videos," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 856–865.

[10] B. Li, W. Zhang, M. Tian, G. Zhai, and X. Wang, "Blindly assess quality of in-the-wild videos via quality-aware pre-training and motion perception," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 5944–5958, Sep. 2022.

[11] H. Wu *et al.*, "FAST-VQA: Efficient end-to-end video quality assessment with fragment sampling," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 538–554.

[12] H. Wu *et al.*, "Neighbourhood representative sampling for efficient end-to-end video quality assessment," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15185–15202, Dec. 2023.

[13] H. Wu *et al.*, "Exploring video quality assessment on user generated contents from aesthetic and technical perspectives," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 20144–20154.

[14] G. Yue *et al.*, "Perceptual quality assessment of enhanced colonoscopy images: A benchmark dataset and an objective method," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 10, pp. 5549–5561, Oct. 2023.

[15] Z. Liu *et al.*, "Video Swin Transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 3202–3211.

[16] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

[17] J. Antkowiak *et al.*, "Final report from the Video Quality Experts Group on the validation of objective models of video quality assessment, March 2000," VQEG, Mar. 2000.

[18] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, 2019.

[19] Y. Dong, X. Liu, Y. Gao, X. Zhou, T. Tan, and G. Zhai, "Light-VQA: A multi-dimensional quality assessment model for low-light video enhancement," 2023, arXiv:2305.09512.